

Transfer learning of ESM3 for enzyme-substrate prediction

Nate Smith (nss97), Alen Zimic (amz63), Chase Holdener (ch2228), Darke Hull (drh257)

Repository: <https://github.com/alenzimic/Transfer-Learning-of-ESM3-for-Enzyme-Substrate-Prediction>

Introduction: Enzymes catalyze the chemical reactions that produce the diversity of molecules found in living organisms. BAHD acyltransferases (BAHDs) are a large, promiscuous plant enzyme family whose members catalyze acyl-group transfer across a wide range of chemically distinct acceptor and donor substrates (**Fig. 1; Kruse et al., 2022**). Predicting which substrate class(es) a given BAHD acts on is a multi-label classification problem characterized by high class imbalance, limited labeled data, and long input sequences (>400 residues).

In prior work (**Smith et al., 2025**), we developed FuncFetch, an LLM-assisted pipeline that screens published manuscripts and retrieves characterized enzyme-substrate interactions and associated metadata. Using published outputs of this pipeline, we compiled curated datasets for BAHDs (n=366) and UGT glycosyltransferases (UGTs, n=482). Each enzyme's experimentally verified substrates were represented as SMILES strings and clustered into categorical superclass labels using NPClassifier (**Kim et al., 2021**). These labels serve as the prediction targets, with the BAHD dataset used for training & validation and the UGT dataset held out to evaluate cross-family generalizability.

We chose ESM3 (**Hayes et al., 2025**), a 98B-parameter masked generative transformer pretrained on 2.78 billion proteins for our source features. It jointly tokenizes protein sequence, 3D structure, and functional annotation as parallel discrete tracks, processing them through a shared bidirectional encoder (rotary embeddings, SwiGLU, SE(3)-invariant geometric attention). However, ESM3's function-token vocabulary is coarse enough that it assigns the same label to essentially all BAHDs, collapsing substrate-level diversity (**Fig. 2**). In lieu of replicating this result, we take it as our starting point and instead ask whether ESM3's intermediate representations encode finer-grained substrate information that classifiers can recover, and whether the resulting models can yield interpretable, biochemically meaningful signals about which protein regions drive substrate specificity (**Fig. 3**).

Methodology: We repurposed internal ESM3 representations as transfer-learning features for a new, more granular enzyme substrate-prediction task. **Two ESM3 intervention points were evaluated: 1) function logits (FL), the per-residue function-track outputs used by ESM3 to predict functional annotations and 2) the last hidden layer (LL) embeddings, the final hidden states from which all track-specific logits are computed via regression heads (Fig. 2).** BAHD protein structures were loaded as PDB files and passed through ESM3 inference to generate per-residue FL and LL tensors. These were mean-pooled across sequence length to obtain fixed-length protein representations and used as input features for multi-label classification of acceptor and donor substrates. The UGT dataset was reserved as a held-out cross-family test. Results are the average of 5-fold cross validation (CV) unless stated otherwise. BAHD-acceptor-micro-AUPR (BA μ PR) was selected as the primary optimization metric because micro-averaging ensures rare classes contribute proportionally to the score, and AUPR is more discriminating than AUROC under the severe class imbalance present in this dataset. Labels with fewer than 2 examples were dropped.

A Lazypredict-inspired (**Pandala, S. R. 2019**) script screened initial linear, tree-based, and other models, applying 5-fold CV and calculating BA μ PR. The top performing model from screening, a logistic regression, a shallow MLP, and a transformer-based model were then compared via a 15x random parameter search. The highest performing models for FL and LL were run 30 additional times in a refined range to select final configurations. The two models were compared with 5-fold and leave-one-out CV in terms of micro/macro AUPR/AUROC to (optimistically) estimate generalizability. The higher performing model on average was chosen as the final model, retrained on the entire BAHD dataset, and evaluated against the 20 shared acceptors in the UGT dataset.

To assess interpretability, we performed two complementary analyses on the final LL-based logistic regression model and the underlying unpooled ESM3 representations. First, because mean pooling, standardization, and logistic regression are linear operations, each one-vs-rest prediction score could be exactly decomposed into per-residue and per-feature contributions, identifying which positions

along the protein sequence most influence a given substrate prediction. Second, to test whether mean pooling obscured localized biological signals, we directly probed the original per-residue ESM3 hidden-state tensors. Residues were grouped into 50 relative-position bins, and each feature-by-bin activation was treated as a univariate classifier for each substrate label, with AUROC used to measure discriminative power and permutation tests used to assess significance. We visualized the strongest feature-position signals with cross-protein heatmaps comparing positive and negative examples, and we checked protein-length distributions across labels to evaluate whether apparent positional signals could be explained by length confounding rather than substrate-specific representation.

Results and Analysis: The original publication of ESM3 predicted broad InterPro annotations for BAHD and UGT enzymes. This novel implementation predicts meaningful, more granular functions. The most prominent challenge in this implementation is having a sufficiently large dataset to represent rare classes which prevented more complex approaches like fine-tuning the ESM3 encoder without overfitting. More data may have improved the performance of nonlinear models in initial screening (**Fig 4**). The success of donor prediction (**Fig 5**) and modest generalizability of acceptor prediction (**Fig 5**) suggests that the ESM3 hidden states contain untapped prediction potential.

Interpretability analysis showed that the predictive signal present in ESM3 was partially obscured by mean pooling over the residue dimension of the protein embeddings prior to modeling. Examining the per-feature contributions (with mean pooling across residues) produced diffuse explanations: no single feature dominated the prediction, which is consistent with the smoothing effect of mean pooling and L2 regularization. However, direct probing of the ESM3 hidden states across features *and* residues revealed feature-specific regions of local residues with substrate-specific signals that the classifier-level explanations did not expose. When individual feature-by-residue-bin activations were evaluated as univariate classifiers, several single feature by residue bins achieved high AUROC for specific acceptor classes, including phenolic acids and naphthalenes, and cross-protein heatmaps clearly showed consistent activation differences between positive and negative enzymes (**Fig 6**). Visualizing these particularly informative residue regions on the folded protein can identify candidate protein regions important for enzyme activity (**Fig 6**). This suggests that ESM3 encodes information relevant to substrate specificity in spatially localized regions. Protein-length comparisons did not show a consistent label-wide separation, reducing the likelihood that these localized signals were explained solely by length confounding, although alignment-aware structural validation remains a future step.

Reflections: This project showed that ESM3 can be repurposed for a more granular biochemical task than the original objective of its function track. Although ESM3 assigned overly broad functional labels to enzymes, its hidden states contained substrate-specific information that simple classifiers could recover. The main limitation was the size and imbalance of the curated dataset: several labels had only a few positive examples, making 5-fold CV estimates unstable and limiting rare-class performance. This also constrained model complexity, since MLPs, transformer heads, or full ESM3 fine-tuning were more likely to overfit than improve generalization.

Future work should expand the dataset across additional enzyme families, use structure- or alignment-aware residue binning, and evaluate position-aware architectures such as attention pooling, regional probes, or adapter-based fine-tuning. Overall, our results suggest that protein foundation models contain latent substrate-specific information, but recovering it requires task-specific feature extraction rather than relying only on their native output tracks. Fine-tuning the ESM3 encoder on substrate-labeled data, for example via adapter layers or low-rank updates, may evoke clearer substrate-specificity features in the hidden states than the frozen representations used here. On the interpretability side, mapping the high-signal residue positions identified by our probing analysis onto known catalytic motifs and conserved domains would strengthen the evidence that ESM3 has learned biochemically relevant features rather than spurious correlations. Building on that computational framework, future work could also explore the effect of alternative residue segmentation strategies, such as single-residue resolution or k-mer-based partitions, on both prediction accuracy and interpretability.

Figures

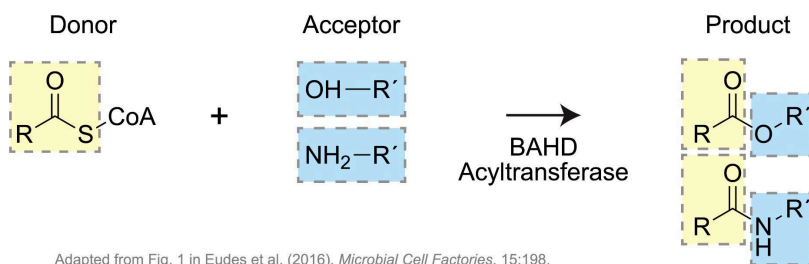


Fig 1: General BAHD acyltransferase reaction scheme. A CoA-thioester donor (yellow) transfers its acyl group to a hydroxyl- or amine-bearing acceptor (blue), forming an ester or amide product. R and R' represent variable chemical scaffolds that differ across BAHD family members and define the substrate classes predicted in this study. Adapted from Eudes et al. (2016).

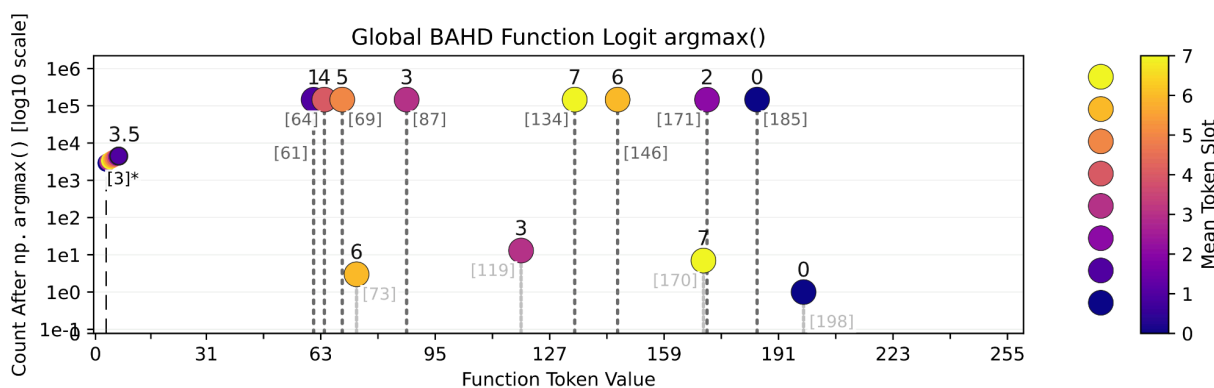


Fig 2: Distribution of ESM3 function tokens predicted across all BAHD residues in the dataset. The 8 tokens highlighted in dark grey brackets correspond to the InterPro annotation output shown in Fig. 2. Nearly all BAHDs receive the same small set of tokens, illustrating the coarseness of ESM3's native functional vocabulary.

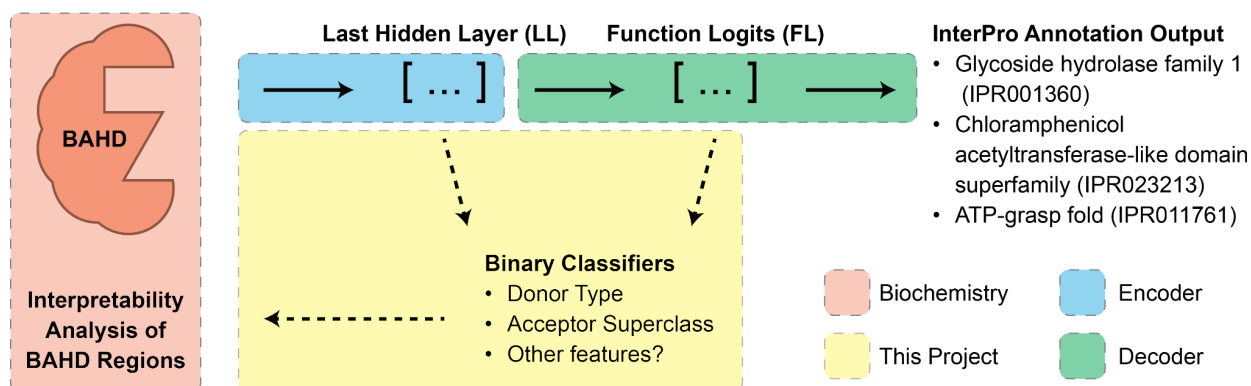


Fig 3: Overview of the project pipeline. BAHD protein structures (PDB files) are passed through ESM3 inference, producing per-residue function logits (FL) and last hidden layer embeddings (LL). These are mean-pooled into fixed-length vectors and used as input features for binary substrate classifiers. The interpretability pathway (dashed arrow, left) decomposes predictions back to per-residue contributions.

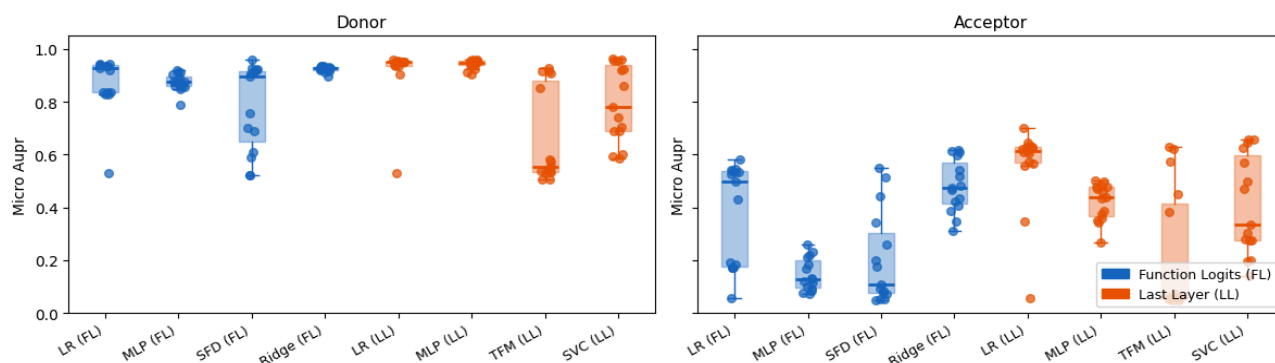


Fig 4: Initial model screening via 15x random hyperparameter search on BAH2 acceptor prediction. Each point represents one 5-fold CV run; y-axis is BA μ PR. FL and LL feature sets are shown separately.

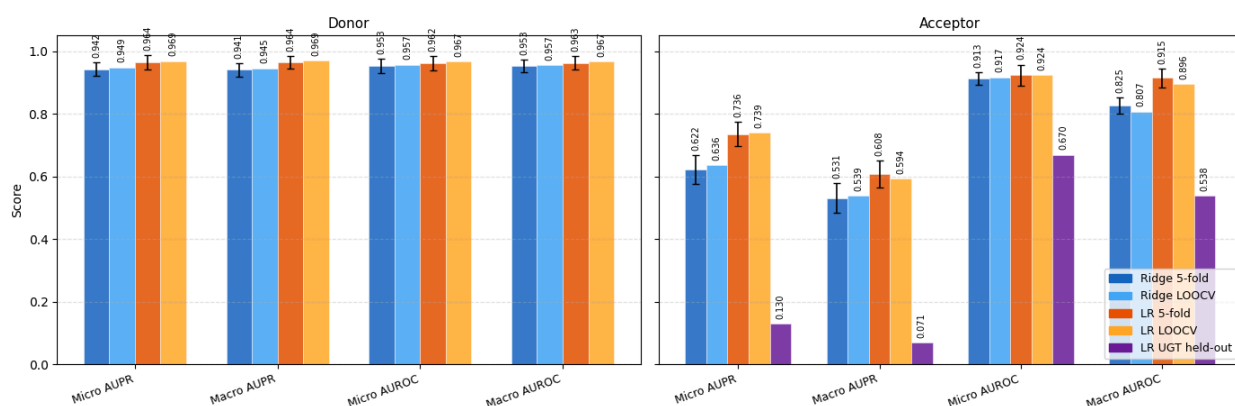


Fig 5: Performance of the final logistic regression model. Left: 5-fold CV on BAH2 train/val set. Right: evaluation on held-out UGT test set (20 shared acceptor classes). Metrics shown include micro/macro AUPR and AUROC.

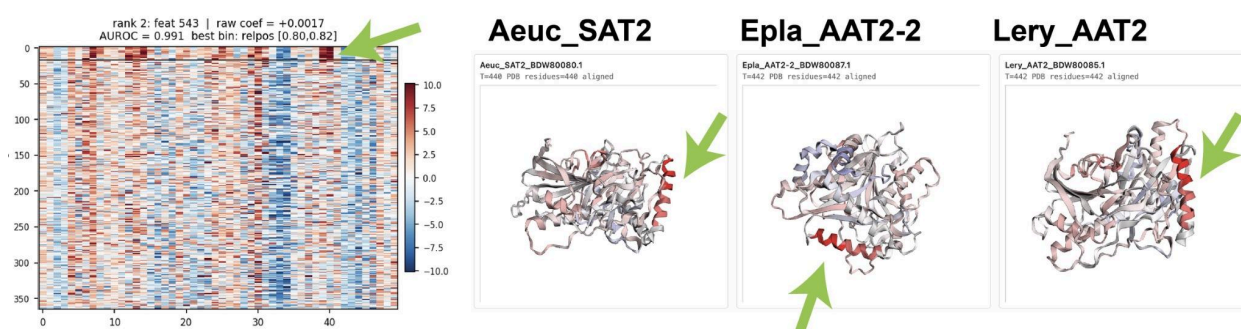


Fig 6: Digging into the raw ESM3 protein embeddings. Left: ESM3 last layer embedding feature 543 out of 1536 at residue bin 40 out of 50 achieves high activation separation between naphthalene acceptors (rows above black line) vs non-acceptors (rows below black line). Right: Visualization of residue bin 40 for 3 naphthalene acceptors Aeuc_SAT2, Lery_AAT2, and Epla_AAT2-2 shows the shared candidate region of the proteins useful for model prediction in naphthalene acceptor behavior.

References

- Hayes, T., Rao, R., Akin, H., Sofroniew, N. J., Oktay, D., Lin, Z., Verkuil, R., Tran, V. Q., Deaton, J., Wiggert, M., Badkundri, R., Shafkat, I., Gong, J., Derry, A., Molina, R. S., Thomas, N., Khan, Y. A., Mishra, C., Kim, C., ... Rives, A. (2025). Simulating 500 million years of evolution with a language model. *Science*, 387(6736), 850–858. <https://doi.org/10.1126/science.ads0018>
- Kim, H. W., Wang, M., Leber, C. A., Nothias, L.-F., Reher, R., Kang, K. B., van der Hooft, J. J. J., Dorrestein, P. C., Gerwick, W. H., & Cottrell, G. W. (2021). NPClassifier: A deep neural network-based structural classification tool for natural products. *Journal of Natural Products*, 84(11), 2795–2807. <https://doi.org/10.1021/acs.jnatprod.1c00399>
- Kruse, L. H., Weigle, A. T., Irfan, M., Martínez-Gómez, J., Chobirko, J. D., Schaffer, J. E., Bennett, A. A., Specht, C. D., Jez, J. M., Shukla, D., & Moghe, G. D. (2022). Orthology-based analysis helps map evolutionary diversification and predict substrate class use of BAHD acyltransferases. *The Plant Journal*, 111(5), 1453–1468. <https://doi.org/10.1111/tbj.15902>
- Pandala, S. R. (2019). LazyPredict (Version 0.3.0) [Python]. <https://github.com/shankarpandala/lazypredict>
- Smith, N., Yuan, X., Melissinos, C., & Moghe, G. D. (2025). FuncFetch: An LLM-assisted workflow enables mining thousands of enzyme–substrate interactions from published manuscripts. *Bioinformatics*, 41(1), btae756. <https://doi.org/10.1093/bioinformatics/btae756>

AI Disclosure

Generative AI was used in this assignment. It was used as a tool for analysis, coding, syntax, and debugging. Additionally, we used AI for paraphrasing and reworking writing.

Definitions (please don't remove points, these are to help with bio/chem vocab)

Enzyme: A protein that catalyzes (accelerates) a specific chemical reaction, acting on one or more substrates and converting them into products.

Substrate: The input molecule(s) an enzyme acts on. In transferase reactions there are generally two substrates: an acceptor (the molecule being modified) and a donor (the molecule supplying the transferred chemical group, including a CoA for BAHDs and UDP-sugar for UGTs).

BAHD acyltransferases (BAHDs): A large plant enzyme family that transfers acyl groups from donor molecules to diverse acceptor scaffolds. Named after the initials of the first four characterized members.

UDP glycosyltransferases (UGTs): A large enzyme family that attaches sugar groups to acceptor molecules, modifying their solubility, stability, and biological activity. Used here as a held-out cross-family test set.

CoA (Coenzyme A): A carrier molecule that activates acyl groups for transfer. In BAHD reactions, the donor substrate is typically a CoA thioester (e.g., acetyl-CoA, coumaroyl-CoA).

SMILES string: A text-based representation of a molecular structure (e.g., CC(=O)O for acetic acid). Used here to encode substrate identity before clustering into categorical labels.